

IAL Physics Unit 4 (WPH14/01) — Complete A* Study Note

Built from analysis of 22 past papers + mark schemes: June 2019 → January 2026

(2019 June and 2019 Oct are the legacy WPH04 “Physics on the Move” papers — 80 marks, 1 h 35. Everything from Jan 2020 onward is WPH14 “Further Mechanics, Fields and Particles” — 90 marks, 1 h 45. The physics overlaps almost completely, so the older papers are still valid practice.)

0. What the paper actually is, and what A* demands

Structure

Paper code	WPH14/01
Time	1 hour 45 minutes
Total	90 marks
Section A	10 multiple-choice questions, 1 mark each
Section B	8–12 structured questions, 80 marks
Asterisked (*) questions	1–2 per paper, always 6 marks, marked by indicative content + structure
Formula sheet	Provided at the back — but it is <i>incomplete</i> (see §7)

Mark-per-minute: 90 marks in 105 minutes $\approx$ **1.15 min/mark**. Section A should take ≤ 12 minutes, leaving ~ 90 minutes for 80 marks.

The three things that separate A* from A on this paper

1. **Command-word discipline.** *Show that, Deduce, Determine, Explain, Discuss, Assess, Suggest, State* each have a different required output. Most lost marks in Section B are lost by producing a correct calculation and then failing to write the one sentence of comparison/conclusion the mark scheme demands. See §6.

2. **Mark-scheme vocabulary.** Examiners award marks for specific phrases: *“change in flux linkage”, “cuts lines of magnetic flux”, “opposes the change that caused it”, “no resultant external force”, “concentration of charge”*. Paraphrases often fail. §8 is a phrase bank — memorise it.
3. **Two-dimensional momentum and exponential capacitor work done fast and cleanly.** These two topics carry the biggest single blocks of marks in nearly every paper and are where time is lost.

Topic weighting observed across the 22 papers

Topic area	Typical marks per paper	Appears
Momentum, impulse, 2D collisions	12–18	Every paper
Capacitors (incl. exponential charge/discharge)	9–18	Every paper
Electromagnetic induction (Faraday/Lenz)	6–17	Every paper
Electric fields, potential, Coulomb	7–16	Every paper
Circular motion	9–17	Every paper
Particle physics / standard model / conservation laws	5–16	Every paper
Alpha scattering	3–13	~90% of papers
Accelerators (linac / cyclotron)	4–13	~85% of papers
Magnetic force on wires/charges	3–12	Every paper
Relativistic effects (muons etc.)	2–7	~70% of papers
de Broglie / electron diffraction / probing	3–4	~60% of papers

PART A — TOPIC 4: FURTHER MECHANICS

1. Momentum and impulse

1.1 Core definitions and equations

- **Momentum** $p = mv$ — a **vector**. Unit: kg m s^{-1} or N s (both accepted; know both).
- **Impulse** $= F\Delta t = \Delta p = \Delta(mv)$. Base units of impulse = **kg m s^{-1}** (asked directly, June 2023 Q4).
- **Newton's second law in momentum form:** $F = \Delta p / \Delta t = \Delta(mv) / \Delta t$.
- **Kinetic energy of a non-relativistic particle:** $E_k = p^2 / 2m$. Derive it if asked (June 2021 Q12a): $E_k = \frac{1}{2}mv^2$, and $p = mv$ so $v = p/m$, hence $E_k = \frac{1}{2}m(p/m)^2 = p^2 / 2m$.
- **Force–time graph: area under the graph = impulse = change in momentum.** This is asked as MCQ (June 2024 Q8) and as a calculation (Oct 2019 Q11a — count squares / estimate area, then $\Delta p = \text{area}$, then $v = \Delta p / m$ remembering the **sign reversal** if the ball rebounds).

1.2 The rebound trap (appears constantly)

A ball of mass m hits a wall at speed u and rebounds at speed v in the **opposite** direction:

$$\Delta p = m(-v) - m(u) = -m(u + v) \quad \text{magnitude} = m(u+v)$$

- If it rebounds at the **same** speed: $|\Delta p| = 2mv$, so $F = 2mv / \Delta t$.
- If it rebounds at **half** speed (Oct 2025 Q9): $|\Delta p| = mv + mv/2 = 3mv/2$, so $F = 3mv / (2\Delta t)$.
- **Force on the wall** is equal and opposite to the force on the ball (Newton's third law) — the direction asked for is usually the *original direction of travel* (i.e. into the wall).
- Jan 2020 Q3 answer: impulse **on the wall** = $2mv$ (positive, in the ball's original direction). Sign conventions matter — state yours.

1.3 Rate-of-change-of-mass problems (jets, propellers, helicopters)

These recur every couple of sessions (helicopter Jan 2022 Q12; flyboard June 2022 Q11; propeller Oct 2024 Q17a).

$$F = \Delta p / \Delta t = (\text{mass flow rate}) \times v = (\Delta m / \Delta t) \times v$$

Worked pattern (Oct 2024 propeller):

1. Area swept $A = \pi r^2$
2. Distance moved by air in time t : $s = vt$
3. Volume $V = As$
4. Mass $m = \rho V$
5. $p = mv$
6. $F = p/t$

For the **helicopter** explanation (4 marks) the mark scheme wants:

- The blades give the air **downward momentum** / change the momentum of the air
- Force on air = rate of change of momentum of the air (downwards)
- By **Newton's third law** the air exerts an equal upward force on the helicopter
- This upward force **equals the weight**, so the resultant force is zero and the helicopter stays at constant height

1.4 Conservation of momentum — the exact wording

"The total momentum of a system remains constant, provided no resultant external force acts." (Alternative accepted: *"total momentum before = total momentum after, in a closed/isolated system"*.)

Both halves are needed for 2 marks. Jan 2024 Q20a and Jan 2021 Q17c both hinge on this.

The assumption question (Jan 2021 Q17c, 2 marks):

- The assumption is that **no resultant external force** acts
- Because if an external force acted there would be an **extra impulse**, so the change in momentum would be different

Why an air track / air table is used (June 2024 Q15a, 2 marks):

- Friction is **negligible**
- So there is **no resultant external force** on the pucks / they behave as a closed system

2. Two-dimensional collisions — the highest-value skill on the

paper

This appears in **every single paper**, usually worth 8–12 marks. There are three question formats and you must be fluent in all three.

2.1 Format 1 — Resolve into components (most common, and usually fastest)

Choose axes along and perpendicular to the initial direction.

$$\begin{array}{ll}\text{Along original direction:} & p_1 = p_2 \cos \theta_2 + p_3 \cos \theta_3 \\ \text{Perpendicular to it:} & 0 = p_2 \sin \theta_2 - p_3 \sin \theta_3\end{array}$$

Standard worked structure (Oct 2025 Q17a — snow tubes):

$$\begin{array}{l}x: m_A u_A = m_A v_A \cos 33^\circ + m_B v_B \cos \theta \\ y: 0 = m_A v_A \sin 33^\circ - m_B v_B \sin \theta\end{array}$$

Divide y by x:

$$\begin{aligned}\tan \theta &= (m_A v_A \sin 33^\circ) / (m_A u_A - m_A v_A \cos 33^\circ) \\ \theta &= 30.9^\circ \approx 31^\circ \quad \checkmark\end{aligned}$$

Mark-scheme structure for a 4-mark version: (1) use of $p = mv$; (2) applies conservation of momentum; (3) applies trigonometry to find the angle; (4) correct final value.

A* habits:

- Always write the two component equations *explicitly labelled* “along original direction” and “perpendicular to original direction” — this alone often earns 2 marks even if the arithmetic fails.
- Dividing the perpendicular equation by the parallel one eliminates the unknown speed and gives θ directly. Do this rather than solving simultaneously.
- Carry unrounded values; round only at the end.

2.2 Format 2 — Scaled vector diagram (when the question says “you should use a scaled vector diagram”)

Seen in June 2022 Q12c, Oct 2021 Q18e(ii), Oct 2022 Q16b(i), June 2021 Q15a, June 2020 Q17b(ii), Oct 2019 Q15c(iii).

Rules that earn the marks:

- State a **scale** explicitly (e.g. “1 cm = 0.10 N s”).

- Draw the momenta **head-to-tail**, with **arrows**.
- **Label every vector** with its quantity *and* magnitude.
- The initial momentum vector is the **closing side** of the triangle (i.e. the resultant of the final momenta), drawn head-to-tail the other way.
- Then **measure** the unknown and convert with your scale.
- Finish with a comparison sentence: *"The measured resultant is ___, which is (not) equal to the initial momentum of ___, so momentum is (not) conserved."*

Why a scaled vector diagram works (June 2022 Q12b, 2 marks): because momentum is a **vector**, and if momentum is conserved the vector sum of the final momenta must equal the initial momentum — so the vectors form a **closed triangle** (masses are equal here, so velocity vectors can be used directly).

2.3 Format 3 — Decay / explosion from rest

If a stationary particle decays into 2 or 3 particles, the momentum vectors must form a **closed polygon** (total = zero).

- 2 products → they move in **exactly opposite directions** with **equal magnitude** momenta (Oct 2025 Q18b(ii); Jan 2026 Q14 lambda decay).
- 3 products → closed triangle (June 2020 Q17b(ii) radon → polonium → lead; Oct 2021 Q18e(ii) pion → muon + neutrino; Oct 2024 Q18b(iii) $\Sigma^+ \rightarrow \pi^+ + n$).

Oct 2024 Q18b(iii) pattern (6 marks — the hardest single momentum question in the set):

$$\begin{aligned} x: \quad p_{\Sigma} &= p_{\pi} \cos 65^\circ + p_n \cos \theta \\ y: \quad 0 &= p_{\pi} \sin 65^\circ - p_n \sin \theta \end{aligned}$$

Solve for $p_n \cos \theta$ and $p_n \sin \theta$, then $p_n = \sqrt{(\dots)}$ and $\theta = \tan^{-1}(\dots)$. Give **magnitude and direction** — the direction mark is separate.

2.4 Elastic vs inelastic — the deduction question

Definitions (know both exactly):

- **Elastic:** kinetic energy is **conserved** (total E_k before = total E_k after). Momentum is *always* conserved in both.
- **Inelastic:** kinetic energy is **not conserved** — some is transferred to other forms (internal energy, sound).

The method (worth 3–5 marks, appears in ~every paper):

1. Calculate total Ek before ($\frac{1}{2}mv^2$ or $p^2/2m$).
2. Calculate total Ek after — **all** the fragments.
3. **Explicit numerical comparison + conclusion in words.**

"Ek before = 127 J, Ek after = 112 J. Ek is not conserved, so the collision is inelastic."

Mark-scheme note: the final mark is for the **comparison and conclusion**, and it is awarded on **ecf** from your own numbers. Never leave the two numbers sitting there without a sentence.

Shortcut worth knowing: when both masses are equal and one is initially stationary, an **elastic** collision always leaves the two bodies moving at **90° to each other**. June 2024 Q15b(i) tests exactly this (answer: 91°, so approximately but not exactly 90° → slightly inelastic). Jan 2026 Q14 uses the same idea in reverse.

A key trap: "the collision is elastic" is sometimes *given* and you must use it as an extra equation (Jan 2026 Q14b(i): use conservation of Ek *and* momentum to find *u*).

2.5 Glider / air-track qualitative results (June 2019 Q14a)

For glider A (mass m_A , speed u) hitting stationary glider B, elastic:

- $m_B < m_A \rightarrow$ A continues forward at reduced speed; B moves forward faster than A.
- $m_B = m_A \rightarrow$ A stops dead; B moves off with A's original velocity.
- $m_B > m_A \rightarrow$ A **rebounds** (moves backwards); B moves forward slowly.

3. Circular motion

3.1 The equation set

Quantity	Equation	Notes
Angular velocity	$\omega = \theta/t = 2\pi/T = 2\pi f$	rad s ⁻¹
Linear speed	$v = \omega r$	
Period	$T = 2\pi/\omega$	
Centripetal acceleration	$a = v^2/r = r\omega^2 = v\omega$	directed towards the centre
Centripetal force	$F = mv^2/r = mr\omega^2$	

Conversion drills (guaranteed MCQ):

- “ N revolutions per minute” $\rightarrow \omega = N \times 2\pi / 60$
- “ N revolutions in t seconds” $\rightarrow \omega = N \times 2\pi / t$
- degrees $\rightarrow$ radians: $\theta_{\text{rad}} = \theta_{\text{deg}} \times 2\pi/360$

3.2 Deriving $a = v^2/r$ (5 marks — appears in Jan 2026, Jan 2024, Oct 2022, Jan 2022, June 2020)

This is a **guaranteed-marks question if memorised**. Mark scheme requires:

1. **Vector diagram:** the two velocity vectors v_A and v_B (equal magnitude, directions tangential at the two positions) drawn head-to-tail with the third side labelled Δv . Note carefully: the diagram is $v_B + (-v_A) = \Delta v$.
2. **Small-angle statement:** for a small angle θ , $\Delta v/v \approx \theta \approx \sin\theta$ (or: arc $\approx$ chord).
3. **Link θ to time:** $\theta = \omega t$ and $v = r\omega$, so $\theta = vt/r$.
4. $a = \Delta v/t$
5. **Algebra to finish:**

$$\Delta v/v = \theta = vt/r \quad \rightarrow \quad \Delta v = v^2 t/r \quad \rightarrow \quad a = \Delta v/t = v^2/r$$

Write all five steps. Students routinely lose the “small angle” mark and the vector-diagram mark.

3.3 Free-body diagrams (2 marks, very common)

Rules that earn marks:

- Arrows **start on the body** (or on a dot representing it), drawn to roughly correct relative lengths.
- **Label each force with a name**, not a symbol only: *weight, tension, normal contact force, friction, lift, drag*.
- **Do NOT draw “centripetal force”** as a separate arrow. This is the single most common error. The centripetal force is the **resultant** of the real forces, not an extra one.
- Never draw an outward force.

3.4 The standard geometries

(a) Conical pendulum / banked aeroplane / toy aeroplane on a wire / hammer throw

Two forces: weight mg and tension/lift T at angle θ to the vertical.

Vertical: $T \cos\theta = mg$

Horizontal: $T \sin\theta = mv^2/r = mr\omega^2$

Divide: $\tan\theta = v^2/(rg) = r\omega^2/g$

And: $\omega = \sqrt{g / (l \cos\theta)}$ [where $r = l \sin\theta$]

- Oct 2021 Q12b(i) asks you to **derive** $\omega = \sqrt{g/(l \cos\theta)}$. Show both component equations, substitute $r = l \sin\theta$, and cancel.
- Oct 2024 Q17b(ii) asks you to show $\tan\theta = v^2/(rg)$.
- Jan 2026 A Q16b(i) asks for $\tan\theta = 2v^2/(gd)$ for a banked track (note $r = d/2$).

(b) Banked track / banking aeroplane (Oct 2022 Q17, Jan 2023 Q13, Jan 2026 A Q16)

Explanation for "how banking allows horizontal circular flight" (4 marks):

- The **vertical component of the lift force equals the weight**
- so the vertical resultant force is zero and the plane does not accelerate vertically
- The **horizontal component of the lift provides the centripetal force** / acts at 90° to the motion
- so the aeroplane follows a circular path

Then if the wings are **levelled** with the same lift magnitude (Jan 2023 Q13b): the lift is now entirely vertical and **greater than the weight**, so there is a resultant upward force and the aeroplane **accelerates upwards** / gains height.

(c) Vertical circles (June 2023 Q15, Oct 2025 Q16, Oct 2023 Q10, Jan 2026 Q10)

At the TOP: $mv^2/r = mg + N \rightarrow N = mv^2/r - mg$

At the BOTTOM: $mv^2/r = N - mg \rightarrow N = mv^2/r + mg$

(N = normal contact force or tension.)

Key reasoning:

- Since speed is constant, the **magnitude of the centripetal force is constant**.
- Force from the surface is a **maximum at the bottom** and a **minimum at the top**.

- **Minimum speed condition:** the object stays in contact when $N \geq 0$; the limiting case is $N = 0$, so $mg = mv^2/r$ and $v_{\min} = \sqrt{gr}$.

The “water in a bucket” explanation (Oct 2025 Q16b, 3 marks):

- The centripetal (resultant) force is the sum of the weight and the contact force
- The minimum contact force is zero
- At that point $mg = mv^2/r$, which gives the minimum speed

(d) Friction providing centripetal force (turntable / coin on a disc / go-kart)

Maximum speed before slipping: $F_{\text{friction}} \geq mv^2/r = mr\omega^2$

- June 2024 Q13: find max friction from the tilt angle ($F = mg \sin\theta$ at the point of sliding, with the height/hypotenuse giving $\sin\theta$), then find the max radius from $F = mr\omega^2$.
- June 2025 Q15b(i): $F_{\max} = mg/30$, so $mg/30 \geq mr\omega^2 \rightarrow r \leq g/(30\omega^2)$. Note that **m cancels** — this is the point of part (ii): a full glass and an empty glass are **equally likely to slide** because mass appears on both sides.

(e) The centrifugal-force question (Oct 2023 Q14, 4 marks) — a guaranteed “assess the claim” question:

- By **Newton’s first law**, without a force the passenger would continue in a **straight line**; a force is needed to change direction.
- The force on the passenger acts **towards the centre** of the circle (centripetal).
- That inward force is exerted by the **car (seat/door) on the passenger**.
- There is **no outward force** on the passenger, so **the claim is incorrect**. (The sensation is the car pushing inwards / the passenger’s inertia.)

3.5 Angular vs linear velocity on a rotating rigid body

Every point on a rotating rigid body has the **same ω** but $v = \omega r$, so points further from the axis move **faster** (June 2025 Q15a(ii), Jan 2022 Q10).

PART B — TOPIC 5: ELECTRIC AND MAGNETIC FIELDS

4. Electric fields

4.1 Definitions and equations

- **Electric field strength:** $E = F/Q$ — “the force per unit positive charge”. Vector. Units N C^{-1} or V m^{-1} (both accepted).
- **Radial field (point charge / outside a charged sphere):** $E = kQ/r^2 = Q/(4\pi\epsilon_0 r^2)$
- **Uniform field (parallel plates):** $E = V/d$
- **Coulomb’s law:** $F = kQ_1Q_2/r^2 = Q_1Q_2/(4\pi\epsilon_0 r^2)$, with $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Inverse-square reasoning drills (frequent MCQ):

- Charges both doubled, separation doubled $\rightarrow F$ unchanged ($4/4 = 1$). (June 2019 Q5)
- Charges $\times 2$ and $\times 2$, separation $\times 2 \rightarrow F$ unchanged. (June 2020 Q8)
- **Potential** varies as $1/r$, **field** as $1/r^2$ — never confuse these.

4.2 Drawing field lines — the exact marking points

Around a point charge / charged sphere (3 marks, appears in Jan 2022 Q11, Oct 2024 Q12a, June 2023 Q17a, Jan 2026 Q18a):

1. At least **4–6 straight radial lines**, evenly spaced, symmetric
2. **Arrows** on the lines — *outwards* for positive, *inwards* for negative
3. Lines drawn so they **meet the surface/centre radially** (perpendicular to the surface of a sphere) and do not stop short or cross

Between parallel plates (Oct 2019 Q12a, Oct 2021 Q16a, Jan 2025 Q16a, June 2025 Q19a, June 2019 Q16a):

1. **Straight, parallel, equally spaced** lines
2. **Perpendicular to the plates**
3. **Arrows from the positive plate to the negative plate** (*Edge effects — curved lines at the ends — are allowed but never required.*)

Two opposite charges (dipole) (June 2021 Q16b, June 2024 Q17a):

- Lines leave + and enter –, curving
- Field lines never cross

- Line spacing is **smallest between the charges** (field strongest there)

Two like charges: lines diverge, with a **null point** between them.

4.3 Equipotentials

- Equipotentials are **always perpendicular to field lines**.
- **Radial field:** concentric circles. For **equal potential-difference intervals** the circles get **further apart** as r increases (because $V \propto 1/r$). Drawing them equally spaced loses a mark.
- **Uniform field:** straight lines parallel to the plates, **equally spaced**.
- **Dipole:** the equipotential midway between equal and opposite charges is the **straight line of 0 V** perpendicular to the axis (June 2024 Q17a(ii) — label it).
- **No work is done** moving a charge along an equipotential.

Scaled equipotential check (Oct 2022 Q15b(ii)): since $V \propto 1/r$, the radius of the 10 V line should be **4× the radius of the 40 V line**. Measure both and compare.

4.4 Electric potential and energy

- $V = kQ/r = Q/(4\pi\epsilon_0 r)$ — **scalar**, so potentials from several charges **add algebraically** (with sign).
- $W = QV$ (work done moving charge Q through p.d. V)
- $W = Q\Delta V$ between two points; for a charge moved between two radii, $W = Q(V_1 - V_2) = kQq(1/r_1 - 1/r_2)$
- **Electric potential energy** of two point charges: $E_p = kQ_1Q_2/r$ (negative for opposite charges).
- The **area under an E–r graph** from r to infinity = the potential at r (Jan 2025 Q9).
- **Capacitance of an isolated sphere:** $C = 4\pi\epsilon_0 r$, hence $Q = 4\pi\epsilon_0 rV$ (Oct 2023 Q20a, Oct 2024 Q16a(ii), June 2023 Q17b(i), Jan 2025 Q12a).

Sign discipline: if the charge is negative, the work done is negative when it moves away from a positive charge. Mark schemes usually accept magnitude with a correct explanatory statement — but say which.

4.5 Distance of closest approach (alpha scattering) — a near-guaranteed question

Appears in June 2020, Oct 2023, June 2022, Oct 2025, Jan 2021, Jan 2026 A.

Method:

1. Convert E_k from MeV to J ($\times 1.6 \times 10^{-13}$) or from eV ($\times 1.6 \times 10^{-19}$).
2. **All initial kinetic energy converts to electric potential energy at closest approach.**
3. Set $E_k = Q_{\alpha} Q_{\text{nucleus}} / (4\pi\epsilon_0 r)$ with $Q_{\alpha} = 2e$ and $Q_{\text{nucleus}} = Ze$.
4. Rearrange: $r = 2Ze^2 / (4\pi\epsilon_0 E_k)$

Worked (Oct 2025 Q19b, $E_k = 8.5 \times 10^{-13}$ J, gold $Z = 79$):

$$r = (79 \times 1.6 \times 10^{-19}) (2 \times 1.6 \times 10^{-19}) / (4\pi \times 8.85 \times 10^{-12} \times 8.5 \times 10^{-13}) = 4.3 \times 10^{-14}$$

5. If then asked for the **maximum force**, put that r into $F = kQ_1Q_2/r^2$ (≈ 20 N for the above).
6. If asked for **maximum acceleration**, follow with $a = F/m_{\alpha}$ (Jan 2021 Q18a $\rightarrow 4.2 \times 10^{27}$ m s $^{-2}$).

Jan 2021 Q18b twist: an alpha deflected through a *smaller* angle does **not** get as close, so the force is smaller and the maximum acceleration is **less**. (Same initial E_k , but not all of it is converted — it retains kinetic energy at closest approach.)

Assumption to state (Oct 2021 Q11b): the nucleus behaves as a **point charge**; the nucleus remains stationary; the alpha particle is a point charge.

4.6 Charged particles in uniform fields

(a) Acceleration through a p.d. (electron gun):

$$eV = \frac{1}{2}mv^2 \quad \rightarrow \quad v = \sqrt{(2eV/m)}$$

Do this in your sleep. Appears in June 2019, June 2020, Oct 2023, Jan 2025, Oct 2025A, June 2025.

(b) Parabolic deflection between plates (Oct 2023 Q22, Oct 2025 Q21, Jan 2026 Q18):

$$E = V/d \quad F = EQ = VQ/d \quad a = F/m$$

$$\text{Time between plates: } t = L/v_{\text{horizontal}}$$

$$\text{Vertical deflection between plates: } s_1 = \frac{1}{2}at^2$$

$$\text{Vertical velocity leaving plates: } v_y = at$$

$$\text{Deflection from plates to screen: } s_2 = v_y \times t_2$$

$$\text{Total deflection} = s_1 + s_2$$

The 5-mark version asks for the **total**; forgetting s_1 (the parabolic part) is the standard error.

Why parabolic (Oct 2023 Q22a, 3 marks):

- The electric field is **uniform**, so the force on the electron is **constant**
- The force is **perpendicular to the initial velocity**, so the horizontal velocity is constant
- Constant vertical acceleration + constant horizontal velocity → **parabolic path** (like projectile motion)

(c) Millikan's oil-drop experiment (Jan 2021 Q19, Jan 2025 Q16 — recurs regularly)

Stationary drop:

$$QE = mg \quad \rightarrow \quad QV/d = mg \quad \rightarrow \quad Q = mgd/V$$

Then divide by 1.6×10^{-19} and comment on whether it is a whole-number multiple.

Falling at terminal velocity (Stokes' law) to find the radius:

$$6\pi\eta rv = mg = \rho(4/3)\pi r^3 g \quad \rightarrow \quad r = \sqrt[3]{(9\eta v / (2\rho g))}$$

(Derivation asked in Jan 2021 Q19a(i) — show the cancellation.)

Evaluating the results (Jan 2021 Q19b): charges cluster around multiples of 1.6×10^{-19} C, **but** the distribution has spread / some gaps, so the data **supports but does not prove** the quantisation.

Systematic-error reasoning (Jan 2021 Q19c): Millikan's e was too **small**; since $Q \propto r^3 \propto \eta^{1.5}$, his η must have been too **small**.

(d) Velocity selector (Oct 2022 Q18b, Jan 2023 Q17):

$$\begin{aligned} \text{Electric force} &= \text{magnetic force} \\ QE &= BQv \quad \rightarrow \quad v = E/B = V/(Bd) \end{aligned}$$

Derivation asked in Jan 2023 Q17a(i). The magnetic field must be directed so the magnetic force **opposes** the electric force — use Fleming's left hand rule and remember to **reverse the current direction for a negative charge**.

5. Capacitance

5.1 Definitions and energy

- $C = Q/V$ — the farad, $F = C V^{-1}$
- **Base units of the farad:** $kg^{-1} m^{-2} A^2 s^4$ (June 2020 Q4 — worth deriving once from $C = Q/V = Q^2/W$)

- **Energy stored:**

$$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}Q^2/C$$

- **Q–V graph:** the **gradient is C**; the **area under it is the energy stored** (Oct 2022 Q5, June 2025 Q2). Note: on a *V against Q* graph, capacitance is the **inverse of the gradient**.

Why the energy stored is $\frac{1}{2}QV$, not QV (Oct 2021 Q7): the battery does work QV , but the capacitor stores only $\frac{1}{2}QV$ because the p.d. rises from 0 to V as it charges — **half the energy is dissipated in the resistance of the circuit**.

5.2 Combinations (not on the formula sheet — memorise)

Parallel: $C_{\text{total}} = C_1 + C_2$ (same V , charges add)

Series: $1/C_{\text{total}} = 1/C_1 + 1/C_2$ (same Q , p.d.s add)

- Two identical capacitors in **parallel** → total charge **doubles** (June 2020 Q3, Jan 2024 Q7).
- Two identical capacitors in **series** (Jan 2025 Q20d): total C is **halved**, so the total charge is halved — **but each capacitor carries that same charge**, because the charge on capacitors in series is equal. Answer: the charge on each is **half** that on the single capacitor.
- **Charge sharing** (Oct 2019 Q12c): total charge is conserved and the final p.d. is common:

$$Q_{\text{total}} = C_1 V_0 ; \quad V = Q_{\text{total}} / (C_1 + C_2)$$

5.3 Charging and discharging — the exponential core

Discharging (through resistance R):

$$Q = Q_0 e^{(-t/RC)}$$

$$V = V_0 e^{(-t/RC)}$$

$$I = I_0 e^{(-t/RC)}$$

Charging (through R from supply V_0):

$$V_C = V_0 (1 - e^{(-t/RC)})$$

$$V_R = V_0 e^{(-t/RC)}$$

$$Q = Q_0 (1 - e^{(-t/RC)})$$

$$I = I_0 e^{(-t/RC)} \quad \text{with } I_0 = V_0/R$$

Time constant $\tau = RC$ (seconds):

- After $t = RC$, the quantity has fallen to $1/e \approx 37\%$ of its initial value (or risen to 63% when charging).
- **Half-life of discharge:** $t_{1/2} = RC \ln 2$ (Oct 2023 Q9).
- Time for a fraction f to remain: $t = -RC \ln f$ (Oct 2019 Q6).

Log-linear analysis (the A*-level technique, June 2023 Q18a, Jan 2021 Q10, Jan 2026 A Q7):

$$I = I_0 e^{(-t/RC)} \rightarrow \ln I = \ln I_0 - t/RC$$

$$\text{Gradient of } \ln I \text{ against } t = -1/RC \rightarrow C = -1/(\text{gradient} \times R)$$

Be able to derive this in two lines. In the MCQ version the answer is $C = -1/(\text{gradient} \times R)$.

Derivation asked (June 2021 Q17b(ii), June 2022 Q15a(iii)): show $V = V_0 - V_0 e^{(-t/RC)}$. Argument: $V_R = V_0 e^{(-t/RC)}$ (the resistor p.d. decays exponentially), and by Kirchhoff's second law $V_0 = V_C + V_R$, so $V_C = V_0 - V_0 e^{(-t/RC)}$.

5.4 Reading a capacitor graph — three valid routes to C

Given a discharge graph of V (or Q or I) against t :

1. **Time-constant route:** find V_0/e (i.e. $0.37 V_0$), read off the time — that time is RC . Then $C = \tau/R$.
2. **Half-life route:** read the time for V to halve; $C = t_{1/2}/(R \ln 2)$.
3. **Log route:** take several pairs, plot $\ln V$ against t , gradient = $-1/RC$. (Most accurate — and the route the mark scheme prefers on “determine” questions.)
4. **Single-pair route:** substitute one (V, t) pair into $V = V_0 e^{(-t/RC)}$ and solve.
Acceptable but a single point is less reliable — say so if asked to evaluate the method.

Tolerance questions (Oct 2023 Q18a, June 2025 Q17b(i), Oct 2025 Q12, Jan 2024 Q11):

- Find the value from the graph

- Compute the allowed range from the stated tolerance (e.g. $\pm 10\%$ of $470\ \mu\text{F} \rightarrow 423$ to $517\ \mu\text{F}$)
- **State explicitly** whether the value lies inside the range and conclude.
- For a “maximum energy” question, use the **maximum** capacitance ($C \times 1.20$ for $\pm 20\%$) before applying $W = \frac{1}{2}CV^2$.

5.5 Explaining capacitor behaviour (the 4–6 mark written questions)

“Explain the variation of V with time as the capacitor charges” (Oct 2024 Q14, 6-mark * question) — indicative content:

1. When the switch is closed there is a **current in the circuit**
2. The **charge on the capacitor increases**
3. The **p.d. across the capacitor increases** and the p.d. across the resistor **decreases**
4. Since $I = V_R/R$, the **current decreases**
5. As the current decreases, the **rate of charging decreases** (so the graph’s gradient falls)
6. Eventually $V_C = V_0$, the **current is zero** and there is no further change

“Explain how the current varies” (Jan 2025 Q20a(i), 4 marks) — same chain, ending with current $\rightarrow 0$ exponentially.

Sketching the current graph: always an **exponential decay from a maximum at $t = 0$** , and you must **label the maximum value** $I_0 = V_0/R$. This is asked repeatedly (Jan 2024 Q18a(ii), Jan 2025 Q20a(ii), Oct 2025 Q20b, Oct 2022 Q12a).

Effect of changing R or C :

- Second identical resistor in **parallel** $\rightarrow R$ halves $\rightarrow RC$ halves $\rightarrow$ charges **faster** (Jan 2024 Q18b). Note: the **final charge is unchanged** because it depends only on C and V_0 .
- Larger C or larger $R \rightarrow$ larger time constant $\rightarrow$ slower.

Circuit-design faults (Jan 2023 Q16a, June 2024 Q12a, June 2021 Q17a): be able to spot that the voltmeter must be **across the capacitor**, the resistor must be **in series and in the circuit during both charge and discharge**, the ammeter must be **in series**, and a **two-way switch** is needed.

Why a data logger rather than a voltmeter and stopwatch (June 2019 Q17c): the p.d. changes **very rapidly** (in under a second), so human reaction time causes large errors; the data logger records **many readings at a high rate** with better precision and timing.

5.6 Applications you should recognise

- **Defibrillator** (Oct 2022 Q12, Jan 2026 Q15): energy $\frac{1}{2}CV^2$ compared with a requirement; discharge current $I_0 = V_0/R$, then $I = I_0 e^{(-t/RC)}$.
- **Touchscreen** (Oct 2019 Q12, Oct 2025 Q20): charge sharing / discharge time.
- **Timing circuit / integrated circuit input** (June 2022 Q15b): find the C that gives a specified V_C at a specified time, then pick from a list of available values.
- **Smoothing an a.c. output** (June 2025 Q17): the capacitor discharges slowly through a large R, so the output is smoothed; with a **low-resistance** bulb, RC is small so the capacitor discharges almost immediately and the output returns to the original shape.
- **Maintaining power during a brief disconnection** (Oct 2021 Q14a): the capacitor stores charge and discharges through the load, maintaining p.d. for a time determined by RC .

6. Magnetic fields and forces

6.1 Equations

Force on a current-carrying wire	$F = BIl \sin\theta$
Force on a moving charge	$F = Bqv \sin\theta$
Radius of a charged particle's path	$r = p/(BQ)$
Magnetic flux	$\phi = BA \cos\theta$ (or $BA \sin\theta$ — see below)
Flux linkage	$N\phi$, unit weber-turns (Wb, or Wb-turns)
Faraday's law	$\varepsilon = -d(N\phi)/dt$

Base units of the tesla: $\text{kg A}^{-1} \text{s}^{-2}$ (Oct 2022 Q6). Units of magnetic **flux:** $\text{Wb} = \text{T m}^2$ (Oct 2023 Q4).

6.2 The angle trap — the single most common MCQ error

$F = BIl \sin\theta$ where **θ is the angle between the current and the magnetic field.**

- Wire **perpendicular** to B $\rightarrow \sin 90^\circ = 1 \rightarrow$ maximum force
- Wire **parallel** to B $\rightarrow$ force is **zero**

- In a square coil with the plane parallel to B: sides **parallel to B** feel **no force**; sides **perpendicular** feel BIl (Oct 2019 Q8, Jan 2022 Q7).
- If a diagram gives the angle to the *plane* or to the *vertical*, convert carefully. June 2023 Q10: poles at 35° to the wire $\rightarrow F = BIl \sin 35^\circ$.
- Jan 2025 Q13: field at 20° to the **vertical**, cable horizontal $\rightarrow$ the angle between the cable and the field is 70° , so $F = BIl \sin 70^\circ$.

6.3 Fleming's left-hand rule

First finger = Field, seCond finger = Current, thuMb = Motion (force).

- For a **negative** charge (electron, β^-), the conventional current is **opposite** to the velocity — reverse your second finger.
- For a **positive** charge, current direction = velocity direction.

6.4 The current balance (a favourite 3-marker)

A wire between magnets sitting on a top-pan balance; the reading changes by Δm grams when the current flows.

$$F = \Delta m(\text{kg}) \times g \quad \text{then} \quad B = F/(Il) \quad \text{or} \quad l = F/(BI) \quad \text{or} \quad I = F/(Bl)$$

Appears in Jan 2021 Q8 (MCQ), Oct 2023 Q13, June 2025 Q13. **Convert grams to kilograms and multiply by 9.81** — the classic slip.

The **reaction** explanation: the wire experiences an upward/downward force; by Newton's third law the magnets experience an equal and opposite force, which is what the balance reads.

6.5 Moment on a coil (electric motor)

$$\text{Force on each long side: } F = BIl \times N$$

$$\text{Resultant moment about the axle} = F \times \text{width} \quad (= 2 \times F \times \text{width}/2)$$

Appears in June 2020 Q11 (with a 20° tilt: use the **perpendicular distance** = width $\times \cos 20^\circ$) and Oct 2024 Q15a(ii).

Why the coil rotates: by Fleming's LHR the force on one side is **up** and on the opposite side is **down** (currents are in opposite directions), producing a **couple/moment** about the axle.

6.6 Circular motion in a magnetic field — $r = p/BQ$

Derivation (2 marks, June 2019 Q15b(i), June 2023 Q12a):

The magnetic force provides the centripetal force:

$$Bqv = mv^2/r$$

$$Br = mv/q = p/q$$

$$r = p/(Bq)$$

Write both lines explicitly.

Standard applications:

- Find B from E_k : $E_k \rightarrow p = \sqrt{2mE_k} \rightarrow B = p/(Qr)$ (June 2023 Q12b, Jan 2021 Q15b, Jan 2024 Q22b(ii))
- Relativistic particles: $E = pc$, so $p = E/c$, then $B = p/(Qr)$ (June 2021 Q18c(ii))
- Specific charge: $e/m = v/(Br)$ (Jan 2023 Q17a(iii), Oct 2023 Q22b(iii))
- Mass spectrometer: after a velocity selector all ions have the same v , so $r = mv/(BQ)$ depends only on m/Q — different isotopes have different radii and only one reaches the detector (Oct 2022 Q18c).

Ratio questions (MCQ):

- Alpha vs proton at the **same speed**: $r = mv/BQ$, alpha has 4× the mass and 2× the charge $\rightarrow r_\alpha = 2r_p$ (Jan 2026 Q9, June 2024 Q10).
- Alpha vs proton with the **same energy**: $p = \sqrt{2mE_k}$ — work it through, don't guess.

6.7 Reading particle tracks (a whole family of questions)

Charge: curvature direction + FLHR + known field direction gives the sign of the charge. If two tracks curve in **opposite** directions, the particles have **opposite** charges.

Neutral particles leave NO track — because they do not **ionise** the medium. (Detectors work by the charged particle **ionising** atoms of the gas/liquid; the ions act as nucleation sites for bubbles or droplets.)

Evidence a neutral particle was produced:

- There is a **gap** between the tracks / tracks appear "from nothing"
- The visible tracks are **not collinear** with the incoming particle, so momentum requires an invisible particle

- A **V-shape** appearing in empty space indicates a neutral particle decayed there (Oct 2025 A / June 2025 Q14c, June 2022 Q14b, Jan 2024 Q12)

Direction of travel: a particle **loses energy** (loses momentum) as it travels, so $r = p/BQ$ decreases — the track **spirals inwards**, and the particle travels from the **larger radius toward the smaller radius**.

Magnitude of momentum from a photograph: measure the radius (use a compass or the chord–sagitta construction), **apply the stated scale factor** (e.g. “the image is half full size”, “1 cm represents 10 cm”), then $p = BQr$.

Comparing two tracks: smaller radius $\Rightarrow$ smaller $p/Q \Rightarrow$ (at the same speed) smaller mass or larger charge.

Antihelium vs proton (Jan 2026 A Q14c): antihelium has charge $-2e$ and mass $\sim 4\times$ the proton, so at the same velocity $r = mv/BQ$ gives $r_{\text{antiHe}} = 2 r_p$, and it curves in the **opposite direction** because the charge is negative.

7. Electromagnetic induction

This is the topic where mark-scheme wording matters most. Learn the chain and reproduce it every time.

7.1 The universal explanation chain (memorise verbatim)

1. There is a **change in (magnetic) flux linkage** with the coil (*or: the conductor **cuts lines of magnetic flux** / cuts magnetic field lines*)
2. An **e.m.f. is induced** (Faraday's law: the induced e.m.f. is proportional to the **rate of change of flux linkage**)
3. There is a **complete circuit / conducting path**, so there is an **induced current**
4. The induced current produces its **own magnetic field**
5. By **Lenz's law**, the induced effect **opposes the change that caused it** — producing a **force opposing the motion**
6. (*Consequence in context: the magnet slows / the balance reading increases / kinetic energy is dissipated / a bigger applied force is needed*)

Six statements $\rightarrow$ six indicative content marks on a * question. This exact chain scores full marks on: June 2021 Q14 (magnet in a copper tube), Oct 2022 Q14 (two tubes, one slit), Jan 2023 Q15a (wire between magnets on a balance), Jan 2021 Q16a (tokamak), June 2022

Q18b (eddy-current brake), Oct 2021 Q15b (electromagnetic brake), Jan 2024 Q21a (induction cooker), June 2024 Q18a (spark plug), June 2025 Q18a (wireless charger), Oct 2025 Q22c (rod on rails).

Context-specific closers you must add:

- *Copper tube with a slit (Oct 2022 Q14)*: the slit **reduces the number of paths for the induced current**, so the current, opposing force and time of fall are all **smaller** — the magnet falls **faster** through the slit tube.
- *Balance reading (Jan 2023 Q15a)*: an upward force acts on the wire; by **Newton's third law** there is a downward force on the magnets, so the **balance reading increases**.
- *Eddy-current brake (June 2022 Q18b, Oct 2021 Q15b)*: the induced current dissipates energy as **heat** ($P = I^2 R$), so the kinetic energy of the vehicle decreases; a thicker plate has **lower resistance**, so a larger current, larger opposing force and greater energy loss.
- *Induction cooker (Jan 2024 Q21a)*: alternating current → alternating magnetic field → changing flux linkage with the pan → induced e.m.f. → **circulating (eddy) currents in the pan** → energy dissipated as heat. The egg cooks only where it touches the pan because the **egg is not a conductor**, so no current can circulate in it.
- *Rod reaching constant velocity (Oct 2025 Q22c)*: the induced e.m.f. **opposes the battery e.m.f.**, reducing the current; when the induced e.m.f. equals the supply e.m.f. the current — and hence the force — is zero, so acceleration is zero.
- *Current not rising instantly when a switch closes (June 2024 Q18c)*: the growing current creates a changing flux linkage in the coil, inducing a **back e.m.f.** which by Lenz's law **opposes the increase** in current.

7.2 Calculations

Straight conductor moving through a field:

$$\varepsilon = Blv \quad (\text{from } \varepsilon = \Delta\Phi/\Delta t = B(lv \Delta t)/\Delta t)$$

- *Aeroplane wingtips (Jan 2024 Q16, Oct 2025 Q14)*: use only the **vertical component** of the Earth's field: $\varepsilon = B \sin\theta \times l \times v$ (or $B \cos\theta$, depending on how the angle is defined — read the diagram).
- **Why the e.m.f. is zero at the equator (Oct 2025 Q14b)**: the Earth's field is **horizontal** there, so there is **no vertical component**, so there is **no change in flux linkage** / the wings do not cut field lines.

- Rod on rails (Oct 2023 Q17).

Coil in a changing field:

$$\phi = BA \quad N\phi = NBA \quad \varepsilon = N \, d\phi/dt = NA \, dB/dt$$

- **From a B–t graph:** find the **maximum gradient** (dB/dt), then $\varepsilon_{\text{max}} = N \times A \times \text{gradient}$ (June 2020 Q15b(ii), Jan 2022 Q18b).
- **From a ϕ –t graph:** $\varepsilon_{\text{max}} = N \times (\text{max gradient of } \phi \text{ against } t)$ (June 2023 Q16c).
- Remember to compute A properly: $A = \pi r^2 = \pi (d/2)^2$ for a circular coil.

Rotating coil (a.c. generator):

$$\varepsilon_{\text{max}} = NBA\omega = \phi\omega \quad (\text{where } \phi = NBA \text{ is the maximum flux linkage})$$

Jan 2026 Q17b uses $\varepsilon_{\text{max}} = \phi\omega$ directly with a rectangular coil of given width $\times$ length.

Area under an ε –t graph = change in flux linkage (Oct 2019 Q16b(i)): because $\varepsilon = d(N\phi)/dt$, so $\varepsilon \, dt = d(N\phi)$; integrating (taking the area) gives $\Delta N\phi$.

7.3 The a.c. generator graph

- The output is **sinusoidal**.
- **e.m.f. is maximum when the plane of the coil is parallel to the field** — the sides are cutting flux at the maximum rate / the flux linkage is momentarily zero but changing fastest.
- **e.m.f. is zero when the plane of the coil is perpendicular to the field** — the sides move **parallel to the field lines** and cut no flux / the flux linkage is maximum but momentarily unchanging.
- **Doubling the rotation speed** (Jan 2026 Q17a(ii), Oct 2019 Q16c): the **frequency doubles** (period halves) *and* the **peak e.m.f. doubles** (because ω doubles). Both changes must be shown/stated.
- **Starting orientation matters:** if the coil starts with its plane **parallel** to the field, the graph starts at **maximum** e.m.f.; if perpendicular, it starts at zero.
- **Connecting a lamp increases the force needed to turn it** (Oct 2019 Q16d): a current now flows; by Lenz's law the induced current produces a force opposing the rotation, so more work must be done.

7.4 The transformer / two-coil chain

For “explain how a p.d. across the input coil causes a current/charge in the second circuit” (June 2020 Q15b(i), June 2019 Q17d, June 2025 Q18a, Oct 2025 Q6):

1. The alternating p.d. produces an **alternating current** in the primary
2. This produces an **alternating (changing) magnetic field** in the core
3. There is a **change of flux linkage** with the secondary coil
4. An **e.m.f. is induced** in the secondary
5. The circuit is complete, so there is a **current**
6. *(if a diode is present)* the diode gives a **one-directional (direct) current** which charges the capacitor

Factors increasing the induced e.m.f.: more turns on the secondary, coils closer together, higher a.c. **frequency** (larger rate of change), larger flux density, larger area, iron core. **Why the secondary p.d. is less than expected (June 2025 Q18b(ii)):** not all the flux produced by the first coil links with the second coil (flux leakage).

7.5 The search coil

- Placed with its axis **parallel to B**, the flux linkage is a maximum; rotated 90° to the perpendicular position, the flux linkage falls to **zero**.
- A p.d. is induced because there is a **change of flux linkage** as it rotates.
- $\phi = BA$; $\varepsilon_{\text{max}} = N \times (\text{maximum gradient of the } \phi\text{-}t \text{ graph})$.

PART C — TOPIC 6: NUCLEAR AND PARTICLE PHYSICS

8. Alpha particle scattering and the nuclear atom

This appears in almost every paper, either as a 1-mark MCQ (“which is **not** a valid conclusion?”) or as a 5–6 mark description.

8.1 The three observation–conclusion pairs (learn as pairs)

Observation	Conclusion
The vast majority of alpha particles pass straight through, undeflected	The atom is mostly empty space
Some are deflected through small angles	There is a concentration of (positive) charge in the atom (repelling the alphas)
A very small number are deflected through more than 90° / come straight back	(Almost all) the mass of the atom is concentrated in a very small volume — the nucleus — and the charge concentration is large

8.2 The MCQ trap you must never fall for

“The nucleus is made up of protons and neutrons” is NOT a conclusion of alpha scattering. The neutron was not discovered until 1932 (Chadwick), long after the 1911 experiments. This distractor appears in June 2023 Q8, Jan 2022 Q8, June 2024 Q7, Oct 2023 Q5, Jan 2026 Q7 and Oct 2022 Q2. Also invalid: “the nucleus is positively charged” (the scattering alone cannot determine the **sign** — see below); “electrons orbit in energy levels”.

8.3 The sign-of-charge subtlety (June 2020 Q14b)

Rutherford’s own point: the deductions are independent of whether the central charge is positive or negative. A **negative** central charge would **attract** the alpha particle, and a hyperbolic attractive orbit can also deflect it through a large angle (the alpha swings round the far side). The scattering **distribution** is the same for attraction and repulsion, so the experiment cannot distinguish them.

8.4 Supporting details examiners ask for

- **Why the foil must be very thin** (Oct 2021 Q13b): so alphas pass through **few layers of atoms** and are unlikely to undergo **multiple deflections** or be **absorbed** — alpha particles are strongly ionising and have very low penetration.
- **Why a vacuum** is needed: alphas would be absorbed / scattered by air molecules (alphas are strongly ionising and have a short range in air).
- **Why alpha, not beta or gamma** (Oct 2022 Q15a): beta particles are far **less massive**, so they would be deflected by the atomic **electrons** as well; beta is more penetrating and less likely to be scattered. Gamma is **uncharged**, so it experiences **no electrostatic force** and would not be deflected at all. Alpha has **double the charge** so experiences a larger force at the same separation.

- **Why the detector is movable / the source is in a lead box:** to measure the number scattered at different angles; the lead collimates the beam and shields the experimenters.
 - **What Rutherford's quote means** (Oct 2021 Q13a): the pre-existing **plum-pudding (Thomson) model** had charge and mass **spread evenly**, which would produce only small deflections; the large-angle scattering was therefore astonishing and required a small, dense, charged nucleus.
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9. Electron beams and thermionic emission

9.1 Thermionic emission (a 1-mark gift that appears in most papers)

Thermionic emission: the filament is heated by an electric current; electrons in the metal **gain kinetic energy**; some gain enough energy to **escape from the surface** of the metal.

Distractors offered: ionisation, photoelectric effect, annihilation, excitation, beta decay. The answer is always **thermionic emission** when a *heated filament* is mentioned.

To increase the current in a diode/electron gun: **increase the p.d. applied to the heater** (more electrons released per second), or increase the accelerating p.d. (Jan 2026 Q6).

9.2 The electron gun

Work done by field = kinetic energy gained

$$eV = \frac{1}{2}mv^2 \quad v = \sqrt{(2eV/m)}$$

- The electron gun must be in a **vacuum** so electrons are not absorbed/scattered by air molecules.
- The **electronvolt**: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$; $1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J}$.

9.3 Electron diffraction and de Broglie

$$\lambda = h/p \quad p = mv \quad \text{and, combining with } eV = \frac{1}{2}mv^2: \quad \lambda = h/\sqrt{(2meV)}$$

Diffraction pattern reasoning: the rings get **smaller** when λ gets **smaller**, i.e. when the momentum/energy of the electrons **increases** (June 2019 Q9, June 2021 Q5). Larger atomic spacing also gives smaller angles.

What the pattern proved (Jan 2023 Q17b): electrons produce a **diffraction pattern**, which is a **wave property**; therefore electrons — previously regarded purely as particles — exhibit **wave–particle duality**.

Why high energies are needed to probe nucleons (June 2024 Q16c, and MCQ in nearly every paper — 3 marks):

1. High energy $\Rightarrow$ **large momentum**
2. $\lambda = h/p$, so the **de Broglie wavelength is very small** — comparable to the diameter of a nucleon
3. This gives **better resolution** / more detailed diffraction information about the structure

MCQ trap: “to produce individual quarks” is **wrong** — quarks cannot be isolated (quark confinement). “To ionise the nucleons” is wrong. The correct reasons are (a) short de Broglie wavelength and (b) sufficient energy to create new particles.

10. Particle accelerators

10.1 Linear accelerator (linac)

How it works (6-mark * question — indicative content from June 2024 Q16b, June 2022 Q13a, Oct 2024 Q16b(ii), June 2025 Q16a, June 2023 Q13a):

1. Charged particles are **accelerated by the electric field in the gaps between the drift tubes**
2. There is **no electric field inside a drift tube**, so the particle travels at **constant speed** while inside it (it is shielded)
3. The a.c. supply has a **constant frequency**
4. The **polarity of the p.d. reverses while the particle is inside a tube**, so the field is always in the accelerating direction when the particle reaches the next gap
5. The particle must spend the **same time in each tube** (half a period); since its **speed increases**, $L = v \times T/2$ means the **tubes must get progressively longer**
6. In the final section the particles are travelling **close to the speed of light** and their speed no longer increases appreciably, so the tubes are of **constant length**

MCQ answers to lock in:

- Reason the tubes get longer: *the particles must spend the same time in each tube* (Oct

2023 Q3, Oct 2022 Q8) — **not** because the frequency changes.

- Why equal-length tubes at the end: *the speed of the particles is no longer increasing* (June 2019 Q6).
- "The applied p.d. changes polarity when the particles are **inside the tubes**" (Jan 2026 Q8).

10.2 Cyclotron

How it works (6-mark * question — Jan 2025 Q17c, Jan 2021 Q15a, Jan 2024 Q22b(i), Oct 2019 Q14a):

1. The alternating p.d. between the dees creates an **electric field in the gap**
2. The **electric field accelerates** the charged particles as they cross the gap, so they gain kinetic energy
3. Inside the dees there is **no electric field**; the **magnetic field** exerts a force **perpendicular to the velocity**
4. This provides the **centripetal force**, so the particle follows a **circular path** inside each dee
5. The **polarity of the p.d. reverses every half cycle** (while the particle is inside a dee)
6. So the field always accelerates the particle across the gap; the radius increases each time ($r = mv/Bq$), producing a **spiral of increasing radius**, until the particle leaves at the maximum radius

Key equations:

$f = Bq/(2\pi m)$ (frequency of the alternating p.d. — the cyclotron frequency)

$T = 1/f = 2\pi m/(Bq)$ Time in ONE dee = $T/2 = \pi m/(Bq)$

At the exit: $p = BQr_{\text{max}}$, $E_k = (BQr)^2/(2m)$

Note the frequency is **independent of the speed and radius** — this is why a constant-frequency supply works (a favourite MCQ, Oct 2025 Q8, Jan 2023 Q7).

MCQ points:

- To increase the energy gained per revolution: **increase the maximum p.d. across the dees** (Jan 2022 Q9).
- As the radius increases, the **velocity increases** (June 2022 Q8).
- The particles are accelerated by the electric field **between** the dees, not inside them

(Oct 2024 Q8).

- A cyclotron cannot accelerate **neutrons** (uncharged).

10.3 Why high energies / colliding beams

Why high energy is needed to create massive particles (Oct 2022 Q13b, Oct 2019 Q17b, Oct 2025 Q15d):

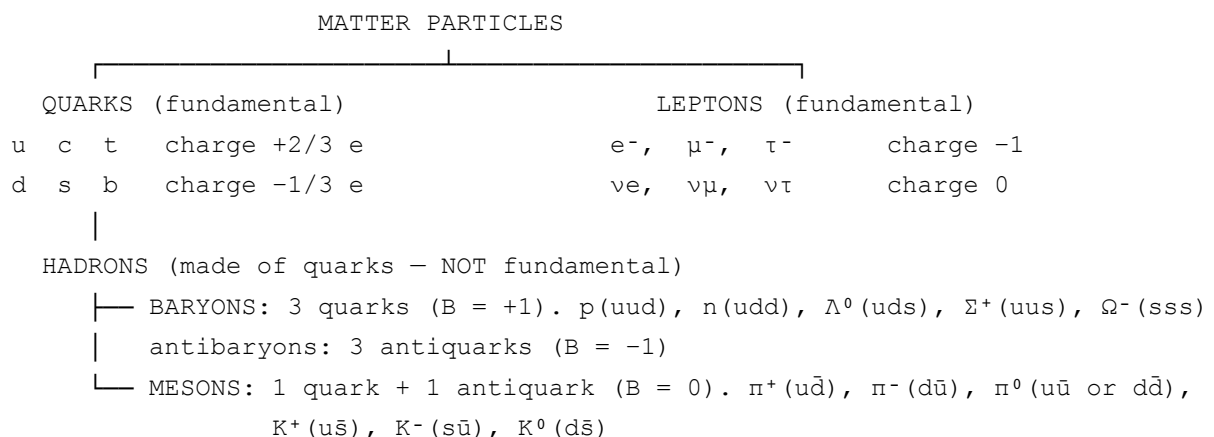
- The new particle has a **large rest mass**
- By $\Delta E = c^2 \Delta m$, energy is required to create that mass
- The **kinetic energy of the colliding particles is converted into the mass** of the new particles

Advantage of colliding beams over a fixed target (Jan 2023 Q18b(i), June 2022 Q13b — 4 marks):

- With two beams of equal and opposite momentum, the **total momentum before the collision is zero**
- Momentum must be conserved, so the products can be created **at rest** — i.e. **all** the kinetic energy is available to be converted into mass
- With a stationary target, the system has net momentum before the collision, so the products **must** carry kinetic energy away, leaving **less energy available for mass**
- Therefore colliding beams can produce particles of **larger mass**

11. The Standard Model

11.1 The classification map (draw this from memory)



Learn to say, in one line each:

- "A baryon is made of three quarks."
- "A meson is made of one quark and one antiquark."
- "Leptons are fundamental particles — they are not made of quarks."

11.2 Quark structures you must be able to derive

Use the charges: u, c, t = $+2/3$; d, s, b = $-1/3$; antiquarks have the opposite sign.

- proton **uud** = $2/3 + 2/3 - 1/3 = +1$ ✓
- neutron **udd** = $2/3 - 1/3 - 1/3 = 0$ ✓
- antiproton **$\bar{u}\bar{u}\bar{d}$** = $-2/3 - 2/3 + 1/3 = -1$ ✓
- $\pi^- = d\bar{u} = -1/3 - 2/3 = -1$ ✓
- $\pi^+ = u\bar{d} = +1$
- $K^- = s\bar{u}$ (must contain a strange quark and have charge -1)
- $\Sigma^+ = uus = 2/3 + 2/3 - 1/3 = +1$, a **baryon**

Pentaquark / tetraquark questions (June 2023 Q11, June 2021 Q4, Oct 2025 Q15b): split the given quarks into a valid **3-quark baryon** and a **quark–antiquark meson**, then compute each charge. Example (June 2023): from c, $\bar{c}$, d, s, u $\rightarrow$ meson **$c\bar{c}$** (charge 0)... but the question requires **equal and opposite** charges, so choose meson **$\bar{c}d$** ($-2/3 - 1/3 = -1$) and baryon **cus** ($+2/3 + 2/3 - 1/3 = +1$).

For a tetraquark of charge $+2e$ made from u, d, s, c and antiquarks (Oct 2025 Q15b): e.g. **$c\bar{s}u\bar{d}$** = $2/3 + 1/3 + 2/3 + 1/3 = +2$ ✓

11.3 Antiparticles

An antiparticle has the **same mass** and the **same magnitude of charge but opposite sign**, and opposite baryon/lepton number.

Positron vs electron (June 2021 Q18a, Jan 2026 Q16a — 3 marks): same **mass**; **opposite (positive) charge**; **opposite lepton number** (-1 instead of $+1$). The track curvature in a magnetic field is in the **opposite direction**.

Trap (Jan 2025 Q4): electron and positron do **not** differ in baryon number — both have $B = 0$. The similarity is *same mass*; the difference is *different charge* or *different lepton number*.

Antiproton (Jan 2022 Q2, Oct 2022 Q1): mass **$+1.67 \times 10^{-27}$ kg** (mass is never negative!),

charge $-1.6 \times 10^{-19} \text{ C}$.

11.4 Annihilation and pair production

- **Annihilation:** particle + antiparticle $\rightarrow$ **two photons** (two, so that momentum is conserved).

Total energy = $2 \times (\text{rest energy}) + \text{kinetic energies}$

Energy per photon (from rest) = mc^2

Then $E = hf$ and $\lambda = hc/E$

Standard question: $e^+ + e^-$ at rest $\rightarrow$ each photon has energy $m_e c^2 = 8.2 \times 10^{-14} \text{ J}$ (0.511 MeV). If they are moving, **add the kinetic energies** before dividing by 2 (June 2021 Q18d, Jan 2026 A Q18d).

- **Pair production:** photon $\rightarrow$ particle + antiparticle. Minimum photon energy = $2mc^2$. For an electron–positron pair: $2 \times 9.11 \times 10^{-31} \times (3 \times 10^8)^2 = 1.64 \times 10^{-13} \text{ J} = 1.02 \text{ MeV}$. Jan 2026 Q16b: a 1.2 MeV photon **is** sufficient ($1.2 > 1.02$). Do the comparison explicitly.
- **Track evidence for pair production** (Jan 2026 Q16c, 6-mark *):
 1. The photon leaves **no track** because it has **no charge**
 2. The tracks are produced by **charged particles**
 3. The two tracks curve in **opposite directions**
 4. So the two particles have **opposite charges**
 5. The **radii of curvature are the same**
 6. Since $r = p/BQ$, equal radii and equal charge magnitude mean equal momenta — so the **masses are the same**, consistent with a particle–antiparticle pair

11.5 Historical / “how do these fit the standard model” questions

Jan 2020 Q12 / Jan 2026 A Q20 (6-mark *): electron, proton, neutron, meson, muon

- Protons and neutrons are **baryons (hadrons)**
- Baryons are made of **three quarks** — so protons and neutrons are **not fundamental**
- **Mesons** are made of a **quark and an antiquark** — also not fundamental
- **Electrons and muons are leptons**
- **Leptons and quarks are fundamental** (not made of anything smaller)

- The muon is a **second-generation** lepton, like a heavy electron

Why the muon is not a meson (Jan 2021 Q14b): muons are **leptons / fundamental particles**, whereas mesons are made of a **quark and an antiquark**.

Zweig's "aces" vs the modern model (June 2020 Q13): the idea of fractionally charged constituents, 3 for baryons and 2 for mesons, and baryon number 1/3 each, all match the modern quark model; but there are **six** quarks (not four), and the number of quarks does match the number of leptons in the sense of three generations of each.

Why a sixth quark was predicted (Jan 2023 Q18a): quarks and leptons appear in **generations/pairs**; there were **six leptons** but only **five quarks**, so **symmetry** suggested a sixth quark must exist.

Chadwick / neutron criticism (June 2024 Q11b): the neutron is **not fundamental** — it is a **baryon made of three quarks (udd)** — so calling it a new fundamental particle is wrong (though it is correct that it is not simply a proton plus an electron).

12. Conservation laws in particle interactions

12.1 The five laws to check

Law	Rule
Charge	Total charge before = total charge after
Baryon number	Baryons +1, antibaryons −1, mesons and leptons 0
Lepton number	Checked separately for each generation : L_e , L_μ , L_τ . Particles +1, antiparticles −1
Energy (mass–energy)	Total energy including rest mass-energy is conserved
Momentum	Total momentum (vector) is conserved

A* discriminator: lepton number must be checked **per family**. $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$ is allowed (L_e : $0 \rightarrow 1 - 1 = 0 \checkmark$; L_μ : $1 \rightarrow 1 \checkmark$). But $\mu^- \rightarrow e^- + \nu_e + \bar{\nu}_\mu$ is **forbidden** (L_e : $0 \rightarrow +2 \times$; L_μ : $1 \rightarrow -1 \times$) — Jan 2023 Q11b.

12.2 Standard decays to know by heart

β^- decay: $n \rightarrow p + e^- + \bar{\nu}_e$ (an ANTI-electron-neutrino)
 β^+ decay: $p \rightarrow n + e^+ + \nu_e$ (an electron-neutrino)
 π^- decay: $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$
 π^+ decay: $\pi^+ \rightarrow \mu^+ + \nu_\mu$
 μ^- decay: $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$

The bar placement in β decay is asked directly almost every session (June 2023 Q9, Jan 2020 Q1, Jan 2026 A Q1). **β^- produces an antineutrino; β^+ produces a neutrino.**

12.3 How to write the answer for full marks

For “deduce whether the conservation of charge and lepton number apply” (typically 4 marks), lay it out as a table or line-by-line:

Charge:	$-1 \rightarrow (-1) + 0 + 0 = -1$	conserved ✓
Lepton (μ):	$+1 \rightarrow 0 + 1 + 0 = +1$	conserved ✓
Lepton (e):	$0 \rightarrow +1 + 0 + (-1) = 0$	conserved ✓

Then a **concluding sentence**: “Both charge and lepton number are conserved, so the decay is possible.” Marks are given for the *numbers* on each side, not just the verdict.

12.4 Nuclear equations

Alpha: $A \rightarrow A-4, Z \rightarrow Z-2$
 Beta $^-$: A unchanged, $Z \rightarrow Z+1$
 Beta $^+$: A unchanged, $Z \rightarrow Z-1$

Balance nucleon number and proton number on both sides. Deuterium bombardment / neutron release questions (Jan 2020 Q11, Oct 2019 Q14b(i), June 2024 Q11a) just require careful arithmetic — count nucleons and protons separately and state the number of neutrons released.

12.5 Energy conservation in decay

For a stationary particle decaying (Jan 2021 Q13a, Jan 2026 A Q14b(ii)):

- The **rest mass-energy of the parent** equals the total **rest mass-energy of the products plus their kinetic energies**
- The parent’s mass is **greater** than the sum of the products’ masses; the mass difference appears as **kinetic energy** via $\Delta E = c^2 \Delta m$
- **Momentum**: the parent was stationary, so the total momentum after must be **zero** — the products move in **opposite directions** with equal-magnitude momenta

13. Mass–energy and particle units

13.1 The conversions

$$\Delta E = c^2 \Delta m$$

$$\text{MeV}/c^2 \rightarrow \text{kg} : \quad m = (E \text{ in MeV} \times 1.6 \times 10^{-13}) / (3.0 \times 10^8)^2$$

$$\text{kg} \rightarrow \text{MeV}/c^2 : \quad E = mc^2 / (1.6 \times 10^{-13})$$

$$\text{GeV}/c^2 \rightarrow \text{kg} : \quad m = (E \text{ in GeV} \times 1.6 \times 10^{-10}) / c^2$$

Useful rest energies to memorise (they save time and enable sanity checks):

- electron: **0.511 MeV** ($9.11 \times 10^{-31} \text{ kg}$)
- proton: **938 MeV $\approx$ 0.94 GeV** ($1.67 \times 10^{-27} \text{ kg}$)
- muon: 106 MeV ($\approx$ 207 electron masses)
- charged pion: $\sim$ 140 MeV

Why GeV/c is a valid unit of momentum (Jan 2022 Q17d(i), 2 marks): because $E = pc$, so $p = E/c$; energy divided by speed has the units of momentum. Equivalently, since mass can be expressed as E/c^2 , momentum mv has units $(E/c^2) \times c = E/c$.

13.2 The recurring comparison questions

Format: “Show that the mass of particle X (given in GeV/c^2) is N times the mass of a proton.”

Method: convert both to kg (or both to GeV/c^2) and divide. Always state the ratio and compare with the claim.

- Oct 2019 Q17a (Ξ_b , $5.9 \text{ GeV}/c^2$ vs proton, $> 6\times$)
- Oct 2022 Q13a (Z boson, $91 \text{ GeV}/c^2$, $\sim 100\times$ proton)
- Jan 2022 Q17c (muon $106 \text{ MeV}/c^2 = 207$ electron masses)
- Oct 2025 Q15c (tetraquark $2.9 \text{ GeV}/c^2$ vs 2 protons $= 1.88 \text{ GeV}/c^2 \rightarrow$ not equal)
- June 2025 Q14b (percentage difference between two stated masses)

Percentage difference: $| \text{value}_1 - \text{value}_2 | / \text{value}_2 \times 100$ — convert to the **same units first**.

13.3 Maximum mass created in a collision

Total available energy (all kinetic + rest) $\rightarrow$ mass

$$m_{\text{max}} = E_{\text{total}} / c^2$$

- Oct 2024 Q10: two 450 GeV beams annihilating produce a particle–antiparticle pair, so each has mass $450 \times 10^9 \times 1.6 \times 10^{-19} / c^2$ — note the **factor of 2** is on the *number of particles*, not the energy per beam.
 - June 2024 Q14b(ii): only 11% of the total collision energy converts to mass $\rightarrow m = 0.11 \times 2 \times 6800 \text{ GeV} / c^2$.
 - Jan 2023 Q18b(ii): total energy 900 GeV each; maximum kinetic energy of one top quark = (total energy) – (rest energy of the quark), converted to joules.
-

14. Relativistic effects

14.1 What is (and is not) required

You are **not** given the Lorentz factor and are **not** expected to calculate γ . What you *are* expected to do:

1. Perform a **classical** calculation (distance = speed $\times$ lifetime, or time = distance/speed)
2. **Compare** it with the observation
3. **Explain the discrepancy using relativistic effects** in words

14.2 The three effects

- **Time dilation:** for a fast-moving particle, the lifetime measured **in the laboratory frame** is **longer** than the lifetime measured at rest (the proper lifetime). So the **rate of decay decreases** and more particles survive.
- **Length contraction (particle's frame):** in the muon's own frame, the distance to the ground is **contracted**, so it can cover it within its normal lifetime. Either explanation is accepted; be consistent about which frame you are in.
- **Relativistic mass increase:** as $v \rightarrow c$, the mass increases; the speed can **approach but never reach** c , no matter how much energy is supplied. This is why linac drift tubes stop lengthening (June 2024 Q16b, Jan 2026 A Q8).

14.3 The standard muon calculation (Jan 2021 Q14a / Jan 2026 A Q15, 5 marks)

Speed = $0.994c$; lifetime at rest = 2.2×10^{-6} s; distance = 1600 m
 Classical distance = $0.994 \times 3.00 \times 10^8 \times 2.2 \times 10^{-6} = 656$ m

Answer structure:

1. The muon's speed is **very close to the speed of light**
2. Without relativistic effects the muon would travel only **656 m**, which is far less than 1600 m
3. So classically almost none should reach the bottom, yet **74% do**
4. Because the muons travel at relativistic speed, their **lifetime in the Earth's frame is dilated** (or: the distance is contracted in their frame)
5. So most muons survive long enough to reach the ground — the observations **are** consistent with relativity

Variants: Oct 2021 Q16c, Oct 2024 Q13, Oct 2023 Q21d, June 2025 Q4, Oct 2025 Q5, Jan 2022 Q17d(iii), Oct 2022 Q13c.

MCQ answers: longer lifetime and larger mass at high speed (June 2022 Q7); increased lifetime *and* decreased rate of decay (Oct 2025 Q5); the reason is $v \approx c$ (June 2025 Q4).

A subtle "assess" question (Jan 2023 Q18c): a top quark (lifetime 10^{-25} s) still cannot form hadrons (which need $\sim 10^{-23}$ s) even at relativistic speed, because the **lifetime in its own frame is unchanged** — the effect is only observed in the laboratory frame, and the factor needed (100×) is implausible.

PART D — CROSS-CUTTING SKILLS

15. Units, base units, scalars and vectors

Expect one MCQ on this in most papers.

SI base units: kg, m, s, A, K, mol, cd. (Ampere is a base unit; coulomb, joule and tesla are **derived** — June 2024 Q1.)

Derived base units worth memorising:

Quantity	Base units

Momentum / impulse	$\text{kg m s}^{-1} (= \text{N s})$
Force (N)	kg m s^{-2}
Energy (J)	$\text{kg m}^2 \text{s}^{-2}$
Planck constant (J s)	$\text{kg m}^2 \text{s}^{-1}$
Capacitance (F)	$\text{kg}^{-1} \text{m}^{-2} \text{A}^2 \text{s}^4$
Magnetic flux density (T)	$\text{kg A}^{-1} \text{s}^{-2}$
Magnetic flux (Wb)	$\text{T m}^2 = \text{kg m}^2 \text{A}^{-1} \text{s}^{-2}$

Scalars: capacitance, electric potential, energy, mass, speed, charge, time. **Vectors:** momentum, impulse, electric field strength, magnetic flux density, centripetal acceleration, force, velocity. Trap: **electric potential is a scalar; electric field strength is a vector** (Jan 2026 Q2). Density (kg m^{-3}) is scalar (Oct 2024 Q2).

16. Graph and data-handling skills

- **Tangents and gradients:** for $\varepsilon = N \frac{d\phi}{dt}$ or $\frac{dB}{dt}$ you must draw and use a **maximum-gradient tangent**. Mark the triangle on the graph.
- **Areas:** area under $F-t$ = impulse; area under $\varepsilon-t$ = change in flux linkage; area under $Q-V$ = energy stored; area under $E-r$ (to infinity) = potential.
- **Log-linear plots:** $\ln y$ against x gives gradient = the exponential constant.
- **Reading a time constant** off an exponential graph: find where the quantity falls to $0.37 \times$ its initial value.
- **Show-that values:** when a question says "show that X is about *value*", carry out the calculation to **at least 2 significant figures more** than the given value and then state your value alongside the target. In later parts, use the **more precise value you calculated**, not the rounded "show that" value (mark schemes often note the difference the rounded value makes).
- **Significant figures:** give answers to 2–3 s.f. unless told otherwise; never quote more s.f. than the least precise datum.
- **Units on every answer line** — the paper explicitly instructs "include units where appropriate", and unit marks are routinely awarded and lost.

17. Experimental/practical reasoning (recurring 2–3 markers)

- **Why use a video and replay frame by frame?** It allows the display to be read at precise, known time intervals; readings can be repeated/checked; it removes human reaction-time error.
- **Why a data logger?** Fast, automatic readings at high frequency for a rapidly changing quantity.
- **Measuring a diameter from a photograph with a ruler held in front (June 2019 Q15c):** there is **parallax error** because the ruler and the beam are at different distances from the camera; the method is quick but the uncertainty is large.
- **Determining an angle experimentally (Jan 2026 Q14a):** record the paths (e.g. with carbon paper / a video from above), draw the two lines, and measure the angle with a protractor.
- **Determining velocity from a video (Oct 2019 Q11b):** measure the distance moved between frames using a scale in the shot, and divide by the known time between frames (frame rate).
- **Repeat and average;** identify **anomalies**; refer to **resolution** of the instrument when asked whether a reading is detectable (e.g. a voltmeter with resolution 0.1 V cannot register 0.07 V — Oct 2024 Q15b(ii)).

18. Command words: exactly what each requires

Command	What the mark scheme wants
State	A single fact or name. No explanation. Do not waste time.
Calculate	Working + final answer + unit.
Show that...	Full working leading to a value; quote your value and compare with the target (" $= 4.28 \times 10^{-14} \text{ m} \approx 4 \times 10^{-14} \text{ m} \checkmark$ "). No conclusion needed beyond that.
Determine	Calculate, usually requiring you to first extract data from a graph/diagram.
Deduce	Calculate and state a reasoned conclusion in a sentence. The conclusion carries a mark of its own.

Explain	Give physics reasons , linked with "because/so/therefore". A description without causation scores zero.
Describe	State what happens, without needing the reason.
Discuss / Assess / Evaluate / Criticise	Consider both sides , apply physics, and reach a judgement . Say explicitly whether the claim/statement/student is correct.
Suggest	A sensible physics-based idea; there is usually more than one acceptable answer.
Derive	Start from a stated principle or given equation and show every algebraic step to the required result.
Compare	Use comparative language ("larger than", "the same as") for both items.

The single highest-yield habit: every *Deduce / Discuss / Assess / Evaluate* answer must end with a sentence of the form

"[my value] is [greater than / less than / equal to] [the target], so [the claim is correct / the collision is inelastic / the design does meet the requirement]."

This final mark is awarded on **error-carried-forward**, so it is free even if your arithmetic went wrong earlier.

19. Mark-scheme phrase bank (learn these verbatim)

Momentum

- "Total momentum remains constant provided **no resultant external force** acts."
- "Elastic means **kinetic energy is conserved**."
- "Friction is negligible so there is **no resultant external force** / the system is closed."

Circular motion

- "The **resultant force acts towards the centre** of the circle."
- "A force is required to change the direction of motion (Newton's first law)."
- "There is **no outward force** on the passenger."
- "The **magnitude of the centripetal force is constant** because the speed is constant."

Electric fields

- "Electric field strength is the **force per unit positive charge**."
- "The field lines show the **direction of the force on a positive charge**."
- "**All the kinetic energy is converted to electric potential energy** at the closest approach."
- "Assume the nucleus behaves as a **point charge**."

Capacitors

- "The p.d. across the capacitor increases, so the p.d. across the resistor decreases, so the **current decreases**."
- "The **time constant** is the time for the charge/p.d. to fall to **1/e (37%)** of its initial value."

Induction

- "There is a **change in (magnetic) flux linkage** with the coil."
- "The conductor **cuts lines of magnetic flux**."
- "An **e.m.f. is induced**."
- "There is a **complete circuit / conducting path**, so there is a current."
- "By **Lenz's law**, the induced e.m.f./current/force **opposes the change that caused it**."
- "By Newton's third law there is an equal and opposite force on the magnet."
- "**Not all the flux** produced by the first coil **links** with the second coil."

Particle physics

- "A **baryon** is made of **three quarks**; a **meson** is made of a **quark and an antiquark**."
- "**Leptons are fundamental** particles."
- "Neutral particles **do not ionise**, so they leave **no track**."
- "High momentum gives a **short de Broglie wavelength**, which gives **better resolution**."
- "Some of the **kinetic energy is converted into the mass** of the new particles ($\Delta E = c^2 \Delta m$)."
- "The **lifetime is dilated** in the laboratory frame because the speed is close to c."

Alpha scattering

- "Most alpha particles pass **straight through undeflected**, so the atom is **mostly empty space**."
- "A few are deflected through **more than 90°**, so there is a **concentration of charge and mass** in a very small volume."

20. Formula sheet vs must-memorise

Given on the paper (Unit 4 section)

$p = mv$ • $E_k = p^2/2m$ • $v = \omega r$ • $T = 2\pi/\omega$ • $F = ma = mv^2/r$ • $a = v^2/r$ • $a = r\omega^2$ • $F = kQ_1Q_2/r^2$ • $E = F/Q$ • $E = kQ/r^2$ • $E = V/d$ • $C = Q/V$ • $W = \frac{1}{2}QV$ • $Q = Q_0e^{(-t/RC)}$ • $F = BIl \sin\theta$ • $F = Bqv \sin\theta$ • $r = p/BQ$ • $\varepsilon = -d(N\phi)/dt$ • $\Delta E = c^2\Delta m$ • $\lambda = h/p$ Plus all Unit 1 and Unit 2 formulae (suvat, $E_k = \frac{1}{2}mv^2$, $\Delta E = mg\Delta h$, $F = k\Delta x$, Stokes' law, $\rho = m/V$, $V = W/Q$, $R = V/I$, $P = I^2R$, $E = hf$, $v = f\lambda$) and the data list (ε_0 , k , e , m_e , m_p , h , c , u , g).

NOT given — you must know these

- $\omega = 2\pi f$ and $\omega = 2\pi/T$ (**only $T = 2\pi/\omega$ is printed**)
- $W = \frac{1}{2}CV^2$ and $W = \frac{1}{2}Q^2/C$ (**only $W = \frac{1}{2}QV$ is printed**)
- $V = V_0e^{(-t/RC)}$ and $I = I_0e^{(-t/RC)}$ (**only $Q = Q_0e^{(-t/RC)}$ is printed**)
- The **charging** equations $V_C = V_0(1 - e^{(-t/RC)})$, $V_R = V_0e^{(-t/RC)}$
- **Time constant** $\tau = RC$; half-life $= RC \ln 2$
- Capacitors in **series and parallel**
- $C = 4\pi\varepsilon_0 r$ for an isolated sphere (*sometimes given in the question stem — but not always*)
- $V = kQ/r$ for the **potential** of a point charge — **the field equation is printed but the potential equation is not**. This is the most commonly forgotten formula on the paper.
- $W = QV$ for work done moving a charge
- $\phi = BA$ and flux linkage $= NBA$
- $\varepsilon = Blv$ for a conductor moving through a field
- $\varepsilon_{\max} = NBA\omega$

- Cyclotron frequency $f = Bq / (2\pi m)$
 - $E = pc$ for relativistic particles (*usually given when needed*)
 - $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ (*printed in the data list*)
 - Quark charges and the standard model classification
 - Base-unit derivations
-

21. The 15 highest-frequency question types — your revision checklist

Tick each only when you can do it **cold, in under the mark-allocation time**:

1. ☐ 2D collision: resolve into components, find speed and angle
2. ☐ Deduce elastic/inelastic with an explicit numerical comparison
3. ☐ Derive $a = v^2/r$ with a vector diagram (5 marks)
4. ☐ Conical pendulum / banked track: $\tan\theta = v^2/rg$ and the full free-body treatment
5. ☐ Vertical circle: maximum/minimum contact force, minimum speed
6. ☐ Closest approach of an alpha particle, then force and acceleration
7. ☐ Draw radial and uniform electric fields with equipotentials
8. ☐ Combine potentials/fields from two point charges (vector for E, scalar for V)
9. ☐ Find C from a discharge graph by three routes; log-linear gradient method
10. ☐ Sketch and label charging/discharging current and p.d. graphs with maximum values
11. ☐ Explain the charging process (6-mark chain)
12. ☐ Full induction chain: flux linkage $\rightarrow$ e.m.f. $\rightarrow$ current $\rightarrow$ Lenz $\rightarrow$ opposing force
13. ☐ Calculate e.m.f. from a B-t or Φ -t graph using the maximum gradient
14. ☐ Explain a linac **and** a cyclotron (6-mark * answers) plus $f = Bq/2\pi m$
15. ☐ Check charge, baryon number and lepton number line by line and conclude
16. ☐ Alpha-scattering observations and conclusions as three matched pairs
17. ☐ Convert $\text{MeV}/c^2 \leftrightarrow \text{kg}$ and compare masses with a stated claim

18. □ Muon relativity: classical distance calculation + relativistic explanation
 19. □ Read particle tracks: charge sign, direction, neutral particles, momentum from radius
 20. □ Base units of impulse, farad, tesla, weber
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22. Exam-day tactics

1. **Section A first, fast.** Aim for 10 minutes. If a question needs more than 90 seconds, mark it and return. Every distractor in the mark scheme has an explicit reason for being wrong — most of these MCQs are testing one specific misconception listed in this note.
2. **Read the mark allocation as an instruction.** A 2-mark “explain” wants exactly two distinct physics statements; a 6-mark * question wants six.
3. **Write on the graph.** Tangents, triangles, read-off lines, the value of $0.37V_0$ — the marks for “evidence of determining the gradient” are awarded from the graph, not the text.
4. **Never leave a “show that” un-concluded** and never use the rounded show-that value if you calculated a better one.
5. **On * questions, use numbered points or short linked paragraphs** with connectives (“this means that...”, “as a result...”, “therefore...”). Two of the six marks are for **structure and sustained reasoning** — a disconnected bullet list caps you at 4.
6. **State the conclusion sentence** on every deduce/discuss/assess. This is the single most commonly dropped mark in the whole paper.
7. **Units and direction.** If the question asks for magnitude *and* direction, there is a separate mark for direction.
8. **Carry unrounded values** through multi-step calculations; round only at the end.
9. **If stuck on an early part, invent a sensible value and continue** — ecf means later parts are still fully available.
10. **Leave 5 minutes** to check units, significant figures, and that every “deduce” has a conclusion.